

Physical AI

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Contents

- What is Physical AI?
- World Foundation Models
- Robotic Foundation Models

What is Physical AI?

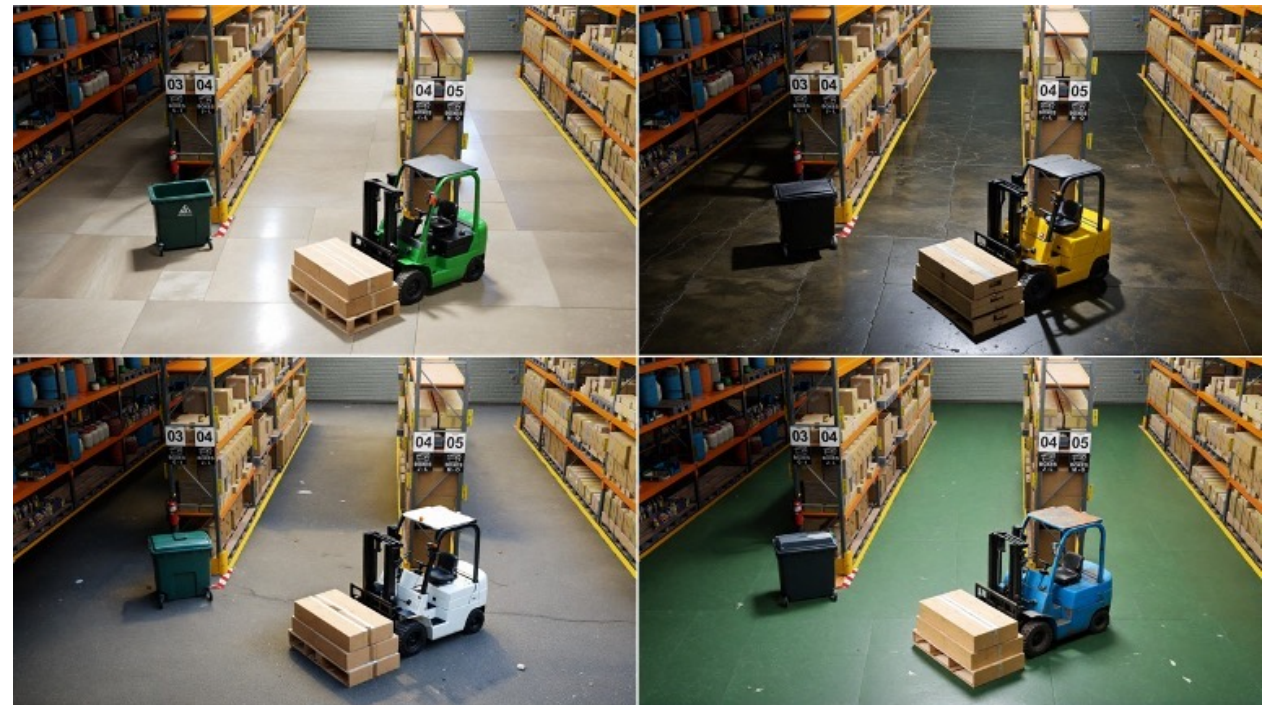
- AI should perceive, understand, and interact with the physical world by using sensors and actuators to gather data and take actions
 - cf. Embodied AI refers to the integration of AI into physical systems, enabling them to interact with the physical world
 - e.g., robots, autonomous vehicles, smart spaces like factories and warehouses, and any network with sensors and actuators in hospital or home scenarios
 - One intrinsic feature is the uncertainty!
 - The uncertainty associated with the acquired information, and incompleteness and uncertainty about the effects of actions over (physical) systems that share the environment with humans

Data Scarcity

- AI must handle diverse, complex situations
 - Autonomous driving: construction zones, weather, parked vehicles
 - Warehouse robotics: lighting, weight, materials, layout



Different types of driving conditions



Different types of flooring and forklift colors

Data Scarcity

- The data required to train for such cases is rarely available
 - Capturing diverse scenes requires physical access, specialized equipment, and manual effort
 - Some environments (e.g., highway construction zones) are dangerous or restricted
- → Most datasets available are limited to common or ideal conditions, falling short of representing the full range of real-world complexity

NVIDIA's Synthetic Dataset Generation

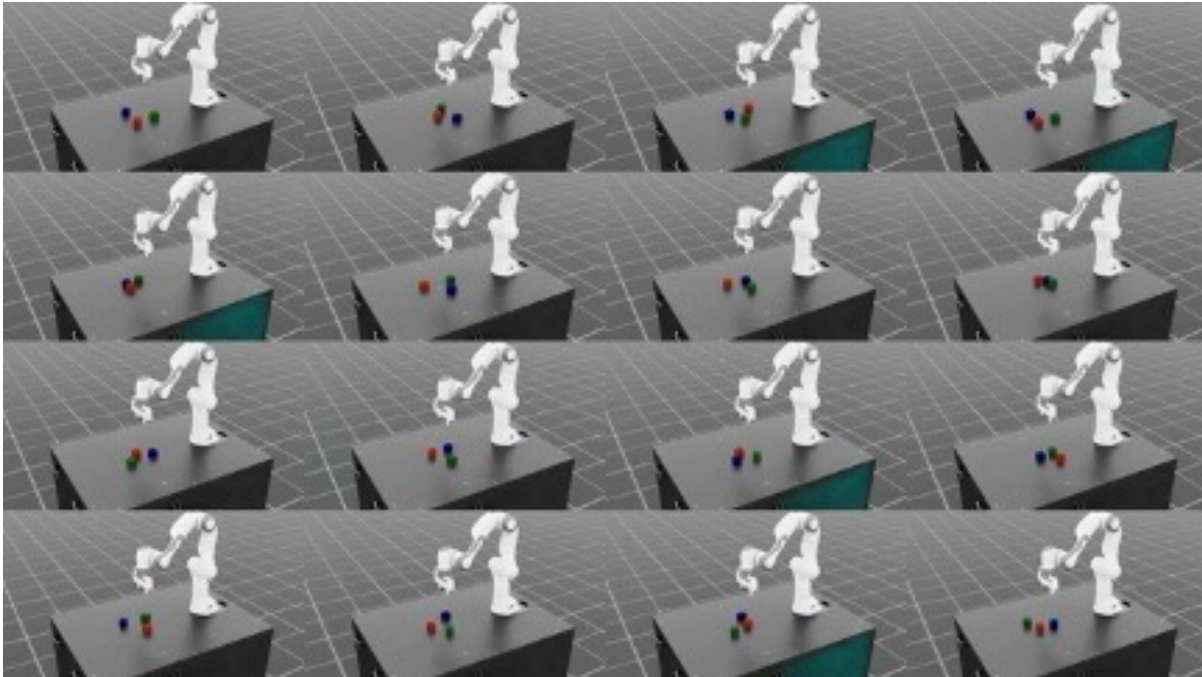
- Generate physics-aware, simulated data at scale
 - Use its high-fidelity graphics engine (e.g., Omniverse, Isaac Sim) to simulate realistic 3D environments that mirror real-world physics, lighting, and object interactions
 - Photorealistic synthetic data in interactive environments like in video games GTA or Overwatch (Lighting, gravity, occlusion, object interaction)
- These synthetic environments enable task-specific training data
 - Realistic lighting and shadow behavior
 - Complex occlusion handling
 - Physically plausible agent-object interaction
 - Scenario diversity (e.g., blocked traffic signals, partial visibility, object manipulation)

NVIDIA Physics AI Ecosystem

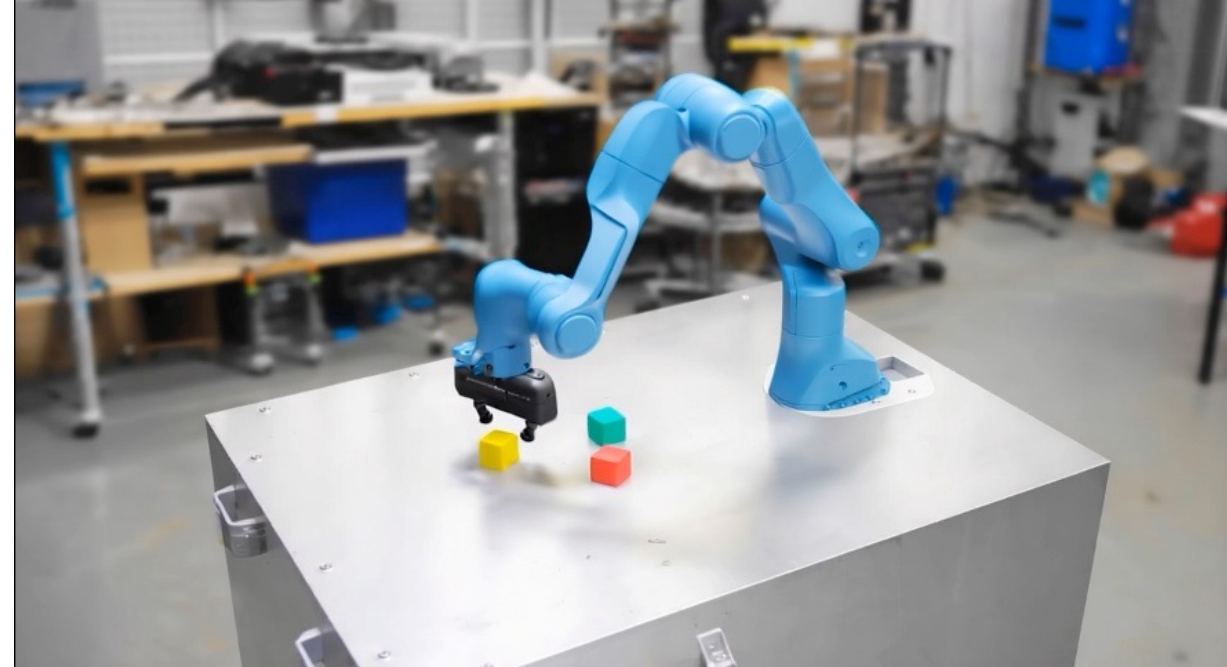
- Omniverse: NVIDIA's simulation engine
 - Photorealistic, physically accurate virtual environments
 - Simulates real-world physics: light reflection & shadows, object collision & gravity
 - Edify (3D asset generation): generate realistic, high-quality 3D content for virtual environments
- Cosmos: world foundation model
 - Trained to understand real-world dynamics across diverse domains
- GRooT: robot foundation model
 - Designed for physical interaction
 - Built on large-scale synthetic trajectory data generated via Cosmos

NVIDIA Omniverse

- A real-time 3D graphics collaboration platform
 - Physics-aware data for training models that generalize well
 - User can vary robot arm material, background environment, and material of boxes



Generate various object placements



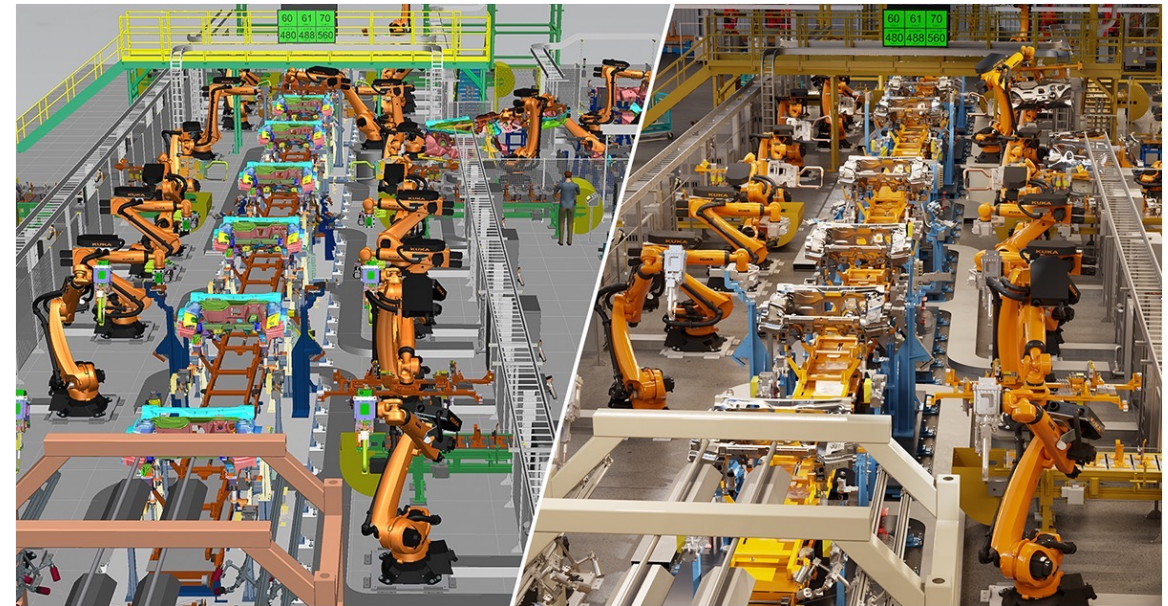
Synthetically generated videos

NVIDIA Omniverse – Digital Twins

- A virtual replica of real-world objects or environments
 - Enable controlled simulation of physical scenarios for AI training
 - Reconstruct environments like homes, warehouses, or objects
 - Can be used to simulate diverse scenarios, and user-interaction results



Reconstructed real objects (its materials)



Reconstructed / Real Farm

Conclusion

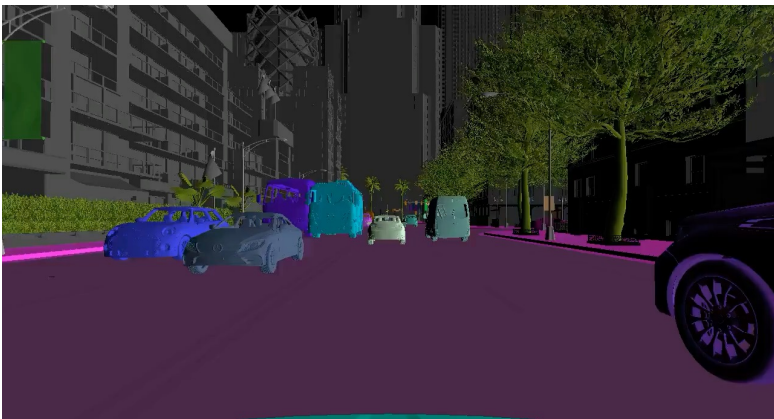
- Synthetic data generation addresses the scarcity of real-world data for training spatial AI models
 - Enables the scalable dataset creation, capturing the complexity and diversity of real-world environments, including rare and extreme edge cases that are difficult to collect in practice
- Impact of NVIDIA Physics AI ecosystem
 - NVIDIA is leading this approach by integrating simulation, pretrained models, and open tools to accelerate development.
 - Creating synthetic datasets can be complex and time-consuming
 - With models trained on high-quality simulations, users can now generate virtual data tailored to their specific tasks

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- **World Foundation Models**
- Robotic Foundation Models

NVIDIA Cosmos

- (Virtual) World Foundation Model (WFM) platform for building physical AI
 - Inputs (text prompts, reference images) can produce videos of 3D objects



Input videos: RGB and segmentation map



Synthesized video

Text Prompt: The camera follows several vehicles driving through a quiet suburban neighborhood on a rainy day



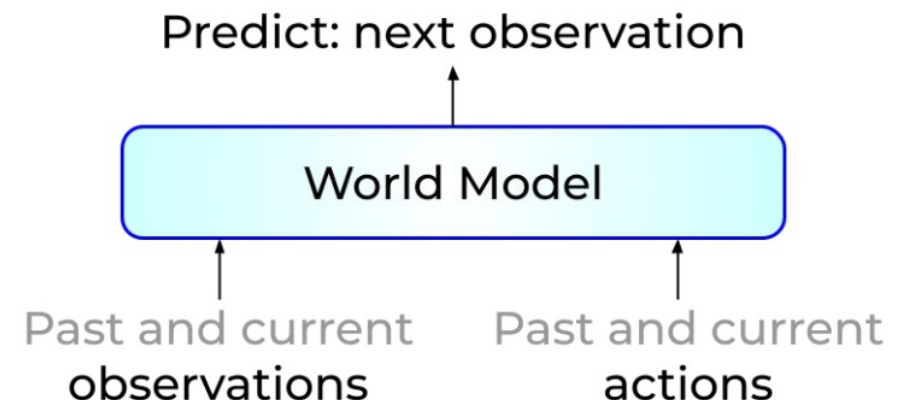
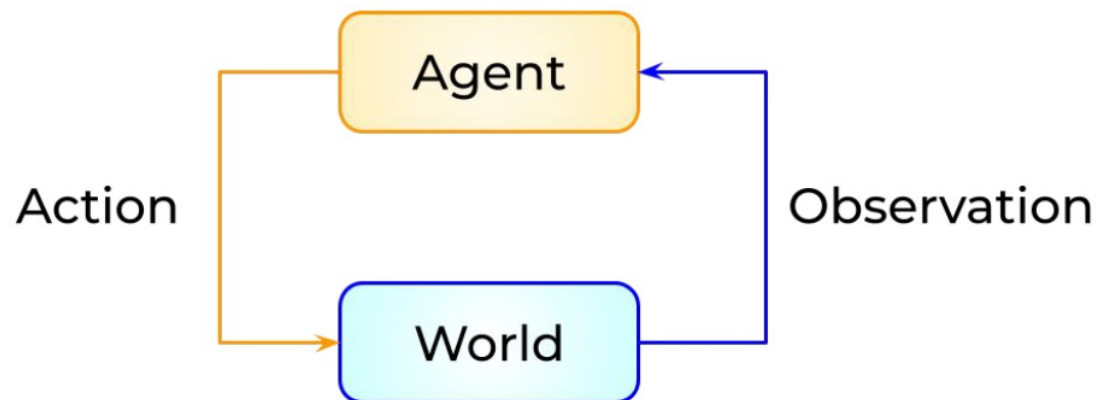
Examples of Autonomous Driving

- Driving datasets can be fully generated in simulation
 - Create rich, scenario-specific training data by digitally twinning the real world,
 - Learn not just basic navigation, but also how to handle rare or complex events such as construction zones, blocked signals, or sudden obstacles



Definition of WFMs

- World models predict the next observation in an environment given the current action and the past observations
 - Reinforcement learning (RL): a machine learning approach where an AI agent learns by interacting with an environment and receiving rewards or penalties based on its actions
 - Over time, it optimizes decision-making to achieve the best possible outcome



NVIDIA Cosmos Flywheel

Curate representative data

Critique, filter and curate training data

Train Downstream Model

Post-train a pre-trained model

Augment synthetic data

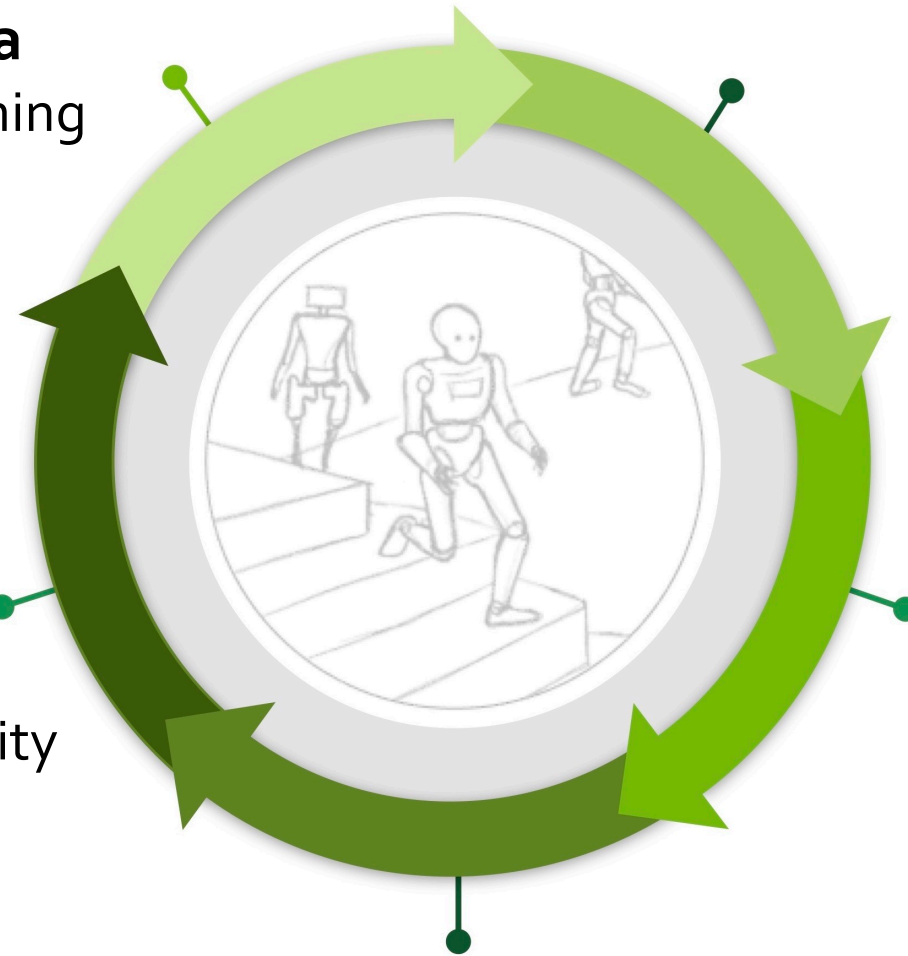
Transferring to photorealism to scale out world state diversity

Evaluate for data gap

Test policy model on real-world tasks and identify poor area of performance

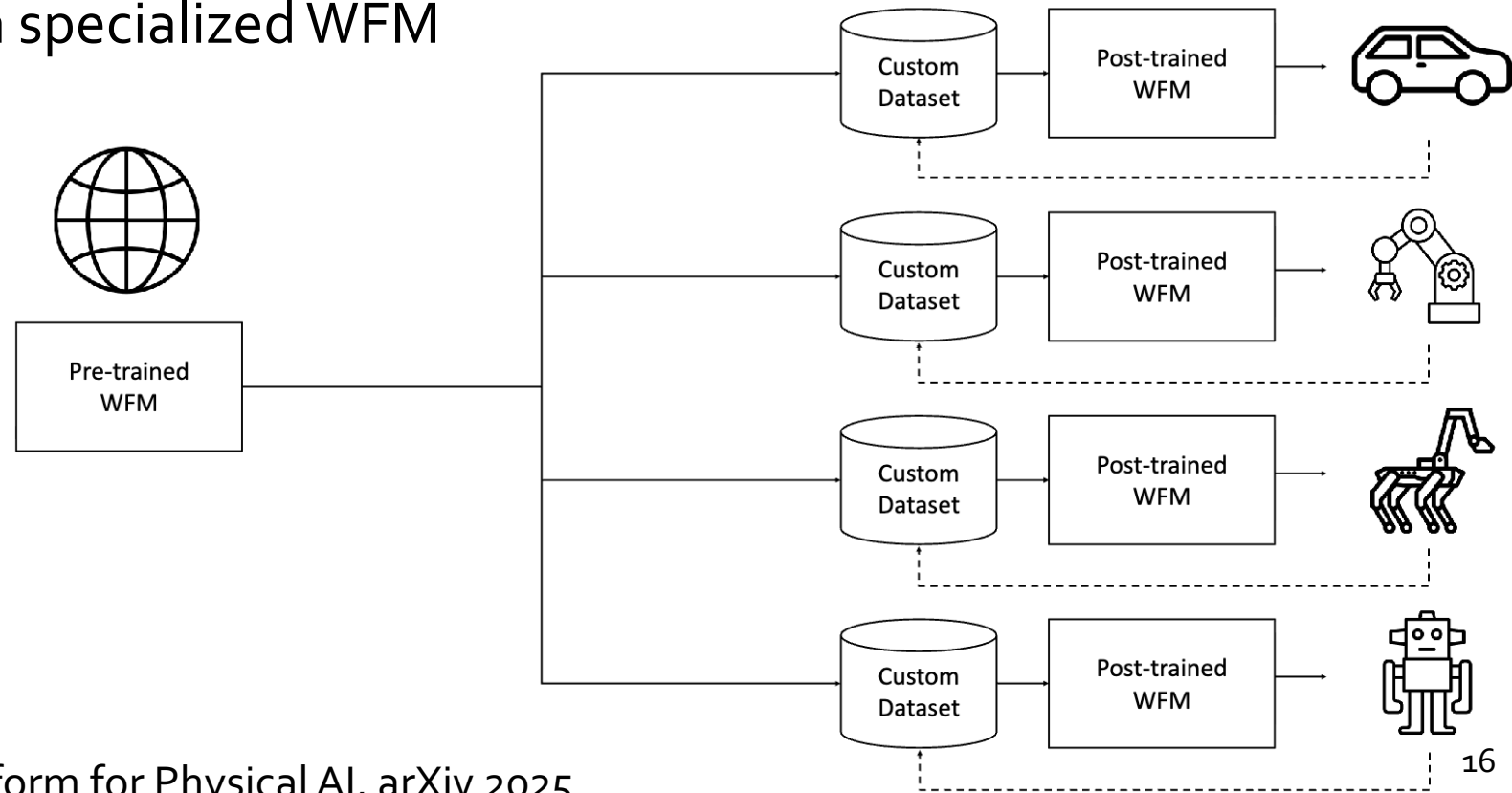
Generate physically accurate data

Simulate 3D ground truth scenarios to generate synthetic data



NVIDIA Cosmos

- A pre-training and-then-post-training paradigm
 - Pretraining: a large-scale video training dataset to expose the model to a diverse set of visual experiences to become a generalist
 - Post-training: fine-tune a specialized WFM using a dataset collected for the targeted, specialized physical AI setup



NVIDIA Cosmos

- An example of GRooT+COSMOS
 - User provided data (limited) → Fine-tune Cosmos → Generate new synthetic data → Robot is trained for a variety of actions with minimal efforts



<https://www.youtube.com/watch?v=8Mwrfvq-GeY>

<https://developer.nvidia.com/blog/enhance-robot-learning-with-synthetic-trajectory-data-generated-by-world-foundation-models/>¹⁷

Examples of Diffusion Models – Google Veo 3

- Higher-quality video generation with audio and speech
 - Released in Google I/O 2025 (5/21/2025)

A medium shot frames an old sailor, his knitted blue sailor hat casting a shadow over his eyes, a thick grey beard obscuring his chin. He holds his pipe in one hand, gesturing with it towards the churning, grey sea beyond the ship's railing. "This ocean, it's a force, a wild, untamed might. And she commands your awe, with every breaking light"



Other World Model – World Labs

- Marble – persistent, navigable, and controllable 3D worlds
 - Given an image or text prompt, generate a 3D world that one can explore — no time limits, no morphing, no inconsistency
 - Better geometry and more diverse styles



Conclusion

- Significant progress towards general-purpose simulators for the physical world but still in the early stages
 - Inconsistent physics especially with complex scenarios like fluid dynamics or complicated material properties
 - Unrealistic movie-style physics
 - Lack of causal understanding with interventional and counterfactual information
 - Substantial memory and computation complexity
 - The availability and quality of data can be a constraint

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Figure AI

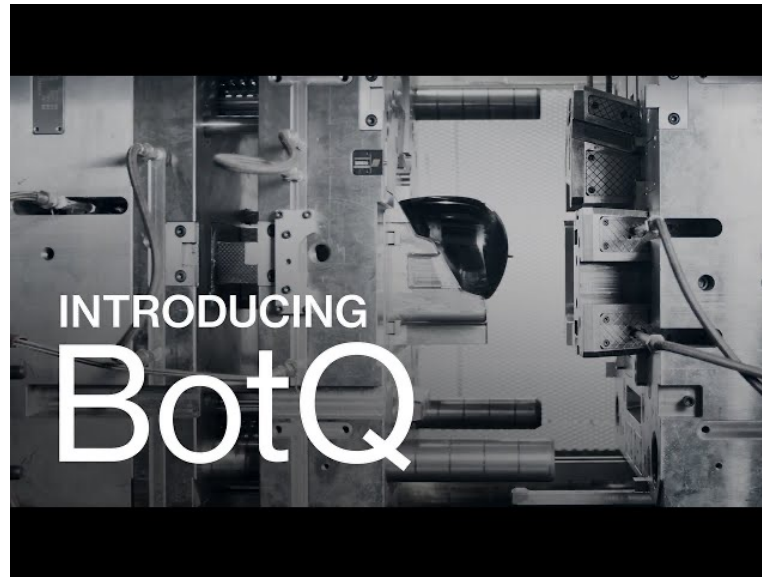
- AI-powered humanoid robots
 - *This week, Figure has passed 5 months running on the BMW X3 body shop production line. We have been running 10 hours per day, every single day of production! It is believed that Figure and BMW are the first in the world to do this with humanoid robots. (2025.10.06)*
– CEO Brett Adcock



https://www.reddit.com/r/singularity/comments/1nzm79t/brett_adcock_this_week_figure_has_passed_5_months/

Figure AI

- Figure 03: a third-generation humanoid (2025.10.9)
 - Designed for home and factory
 - A manufacturing facility BotQ will produce 12,000 Figure 03 humanoids per year (2025.03.15)



Tesla Optimus

- A general-purpose robotic humanoid
 - *Optimus has the potential to be more significant than [Tesla's] vehicle business over time* – Elon Musk (January 2022)
 - Why \$1 trillion pay package for Musk? *\$3 trillion by the end of 2026 as full scale volume production begins of the autonomous and robotics roadmap* (2025.11.7)



Boston Dynamics (Owned by Hyundai Motors Group)

- Shifting from R&D to commercialization
 - Atlas: earliest commercialized bipedal humanoid robot in 2016
 - Focus on selling high-volume robots for industrial and commercial applications like industrial sensing to inspect and monitor facilities (Spot) and warehouse automation (Stretch)



Unitree Robotics



- Pioneering affordability
 - Founded by Wang Xingxing (1990) in May 2016
 - The H2 humanoid robot is priced around \$29,000
 - Unitree Go2 can be purchased in Coupang below 3M KRW



Robotic Foundation Models

- Large-scale AI models designed to serve as versatile building blocks for a wide array of robotics tasks
 - Like LLMs for robots (generalization across diverse robotic applications by extensive training on vast datasets)
 - Unlike traditional model for specific tasks like object manipulation or navigation, foundation models are pre-trained on diverse datasets that encompass a wide range of robotic scenarios
 - Multimodal learning: integrate data from various sensors (e.g., cameras, lidar, and tactile sensors) to understand and respond to their environment holistically
 - Generalization: perform well in tasks they were not explicitly trained on
 - Fine-Tuning: adapt to specific tasks with minimal additional training

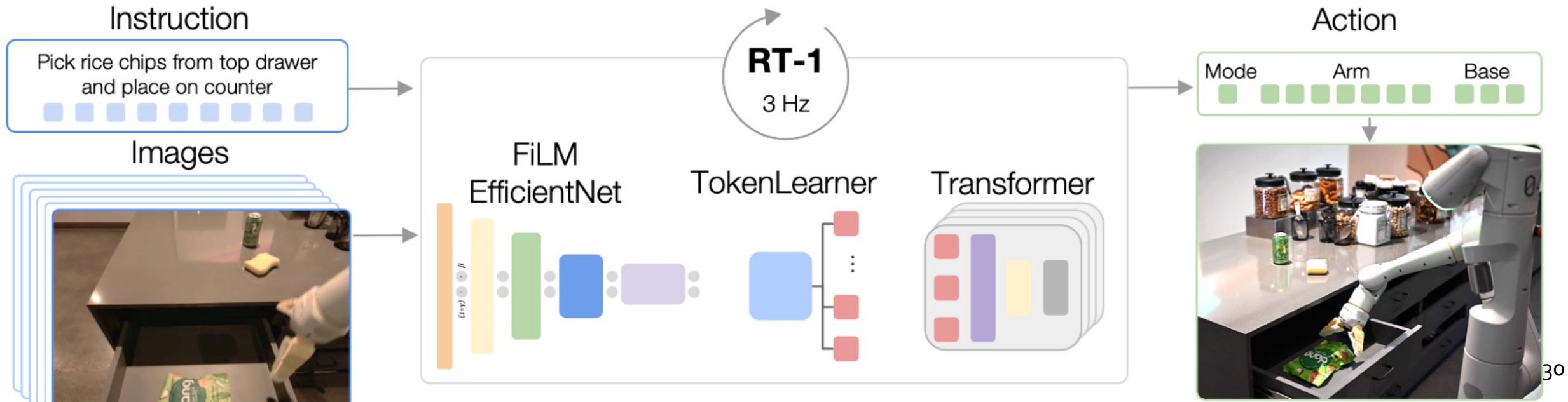
RT-1: Early Work from Google

- A multi-task robotic foundation model
 - A large-scale, real-world robotics dataset of 130k episodes that cover 700+ tasks, collected using a fleet of 13 EDR robots (7 DOF arm, a 2-fingered gripper) over 17 months
 - Significantly improved zero-shot generalization to new tasks, environments and objects



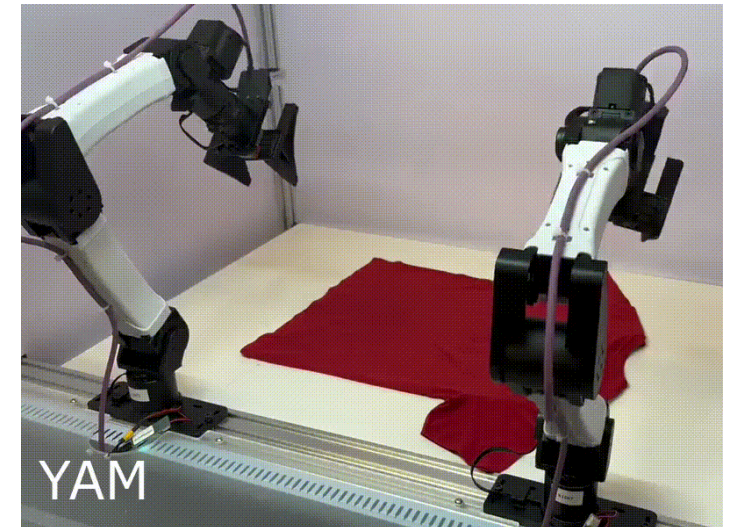
RT-1 Architecture

- Based on conventional LLM architecture
 - EfficientNet-B3 extracts $9 \times 9 \times 512$ spatial feature map and FiLM layers extract task-relevant features cond. on task instruction to 81 tokens
 - Action dimensions: 7 variables for arm movement (x, y, z, roll, pitch, yaw, gripper opening), 3 for base movement (x, y, yaw), and an extra mode variable (controlling arm, controlling base, or terminating the episode)



NVIDIA Isaac GRooT

- Open foundation model for generalized humanoid robot skills
 - NVIDIA Isaac: AI robot development platform (libraries, application frameworks, and AI models)
 - Cross-embodiment model takes multimodal input (language and images, to perform manipulation tasks in diverse environments

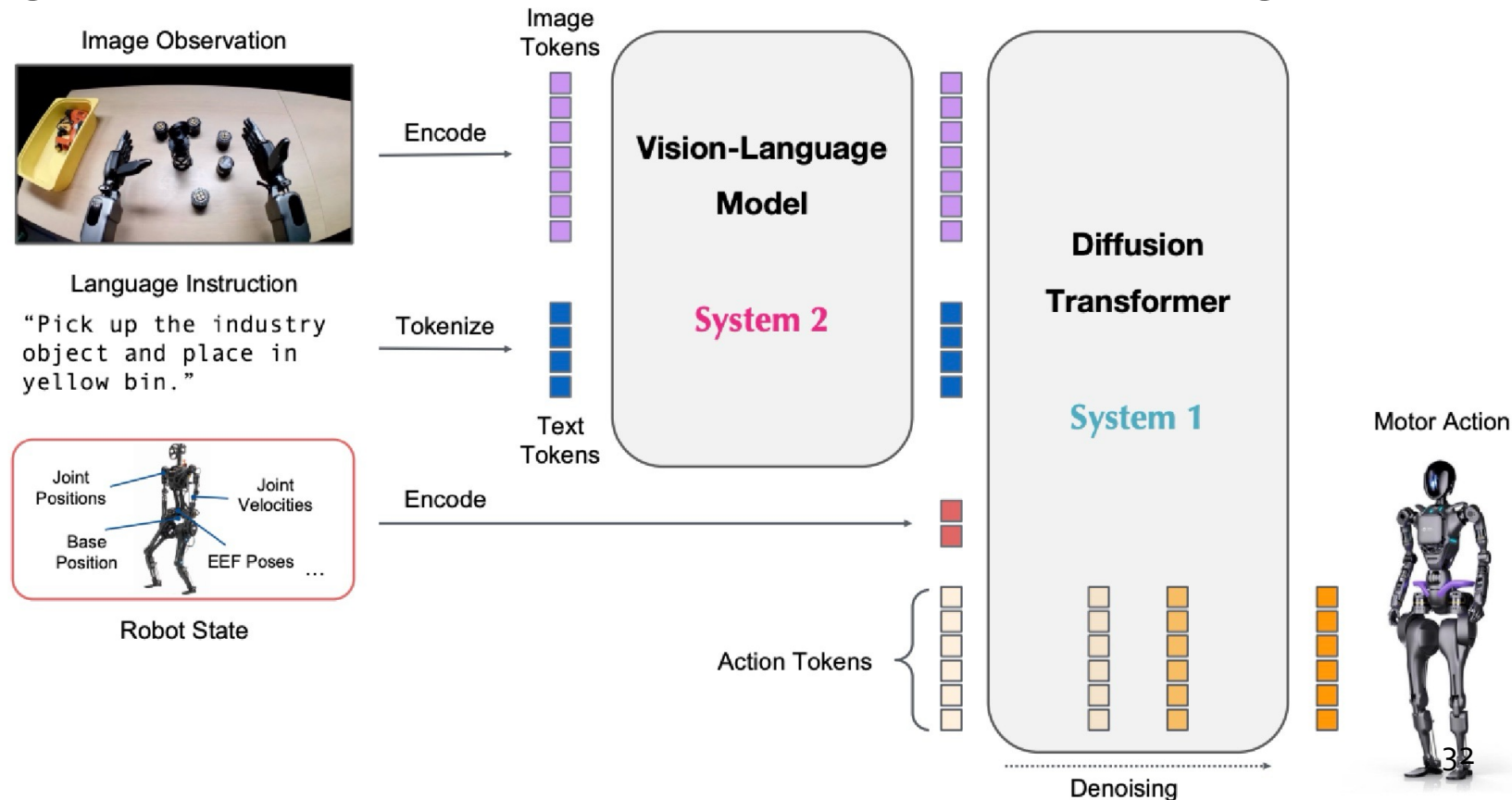


<https://github.com/NVIDIA/Isaac-GRooT>

NVIDIA, GRooT N1: An Open Foundation Model for Generalist Humanoid Robots, arXiv 2025

NVIDIA Isaac GRooT

- A dual-system compositional architecture
 - System 1: a fast-thinking action model, mirroring human reflexes or intuition
 - System 2: a slow-thinking model for deliberate, methodical decision-making
- Inspired by human cognitive processing



Conclusion

- Potential challenges
 - Generalization: robots/tasks are too diverse!
 - Robust real-world deployment: robots should perform reliably in diverse and unpredictable environments
 - Data collection: real robot data should be gathered for each specific robot
 - Recovery from failure: the dataset often contains only successful demos
 - Sequence length limit: this may not solve long horizon tasks
 - Collaborative learning: sharing datasets and models across organizations can accelerate progress
 - Human-robot interaction: how to understand and respond to human cues will be critical for widespread adoption